1. Literature Review on algorithms and tools for visual navigation

1.1. LEVIO: Lightweight Embedded Visual Inertial Odometry for Resource-Constrained Devices (2026)

LEVIO is an open-source, high-efficiency visual-inertial odometry (VIO) system designed specifically for resource-constrained hardware like micro-drones and nano-drones. Traditional state-of-the-art VIO algorithms typically require powerful computing platforms and heavy battery packs that consume watts to tens of watts of power. This heavy hardware exceeds the strict payload limitations of micro-drones, which has historically prevented them from using accurate, infrastructure-less navigation. LEVIO bridges this gap by providing a lightweight 6 Degrees of Freedom (DoF) tracking pipeline that processes 20 frames per second while consuming less than 100 mW on ultra-low-power commercial hardware.
To ensure its effectiveness for drones, the developers benchmarked LEVIO using the EuRoC dataset, which was recorded on board a real drone undergoing highly dynamic movements and varying lighting conditions. This demonstrates that LEVIO can successfully handle the complex, real-world visual tracking challenges that a flying drone would naturally encounter.
Source: https://github.com/ETH-PBL/levio

1.2. Dream to Fly: Model-Based Reinforcement Learning for Vision-Based Drone Flight (2026)

This research paper introduces a new model-based reinforcement learning (MBRL) approach for autonomous drone racing using only raw pixel data. Unlike traditional methods that require complex pre-processing or imitation learning from human experts, the authors utilize the DreamerV3 architecture to train a policy from scratch. By learning a world model to predict future states, the quadrotor successfully navigates race tracks using an end-to-end visuomotor policy. A notable discovery was the emergent perception-aware behavior, where the drone instinctively steered its camera toward visual landmarks without explicit programming. The system was validated through both high-fidelity simulations and real-world hardware-in-the-loop tests. These findings suggest that MBRL provides a more sample-efficient and robust framework for vision-based robotics than previous model-free alternatives.
Source:
https://github.com/NM512/dreamerv3-torch?utm_source=catalyzex.com

1.3. ORB-SLAM3: Unified Visual and Inertial Multi-Map SLAM Library (2023)

This document introduces ORB-SLAM3, an advanced open-source library designed for visual, visual-inertial, and multi-map SLAM across various camera configurations. The researchers highlight two primary innovations: a robust visual-inertial system that utilizes Maximum-a-Posteriori estimation for superior accuracy and a novel multiple map system featuring an improved place recognition method. These advancements allow the system to maintain continuity in challenging environments by creating and merging maps when visual information is temporarily lost. Extensive testing on public datasets demonstrates that ORB-SLAM3 significantly outperforms existing state-of-the-art methods in both precision and reliability. By providing an abstract camera representation, the library supports diverse lens models, such as pin-hole and fisheye, making it a versatile tool for the robotics community. Overall, the work establishes a new benchmark for real-time localization and mapping in both indoor and outdoor scenarios.
Source: https://github.com/UZ-SLAMLab/ORB_SLAM3

1.4. VINS-Fusion: An Optimization-Based Multi-Sensor State Estimator (2023)

VINS-Fusion is an open-source visual-inertial state estimation framework designed for autonomous navigation in GPS-denied environments. The system combines data from cameras and inertial measurement units (IMU) to achieve accurate localization and trajectory estimation. One of the major advantages of VINS-Fusion is its support for multiple sensor configurations, including monocular, stereo, and stereo-plus-IMU systems. The framework performs tightly-coupled nonlinear optimization to improve robustness and reduce accumulated drift during long flights. In drone applications, VINS-Fusion is widely used because it provides stable pose estimation even in environments with rapid motion or temporary visual degradation. Experimental evaluations demonstrated high accuracy and real-time performance on embedded robotic platforms, making it highly suitable for autonomous drones and UAV navigation tasks.
Source: https://github.com/HKUST-Aerial-Robotics/VINS-Fusion

1.5. OpenVSLAM: A Versatile Visual SLAM Framework (2022)

OpenVSLAM is a flexible and open-source visual SLAM framework that supports monocular, stereo, and RGB-D cameras. Unlike many SLAM systems that are designed for specific camera types, OpenVSLAM introduces a camera-model-independent architecture, allowing it to operate with perspective, fisheye, and equirectangular cameras. This flexibility makes the framework especially useful for drone applications, where different camera configurations are commonly used. The system includes modules for tracking,

mapping, loop closure detection, and relocalization while maintaining real-time performance. OpenVSLAM has been successfully applied in robotics, augmented reality, and autonomous aerial navigation. Its modular structure and efficient performance make it an excellent platform for research and experimentation in visual navigation.
Source: https://github.com/xdspacelab/openvslam

1.6. DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras (2023)

DROID-SLAM is a modern deep-learning-based SLAM system that combines neural networks with geometric optimization for highly accurate visual localization and mapping. Unlike traditional SLAM algorithms that rely mainly on handcrafted visual features, DROID-SLAM uses learned feature representations to improve robustness in challenging environments such as low-texture areas, motion blur, and varying illumination conditions. The framework supports monocular, stereo, and RGB-D camera setups and demonstrates state-of-the-art performance on several benchmark datasets. For drone navigation, DROID-SLAM offers improved tracking reliability and trajectory estimation accuracy during dynamic flight scenarios. The integration of deep learning techniques allows the system to generalize better to complex real-world environments compared to classical SLAM approaches.
Source: https://github.com/princeton-vl/DROID-SLAM

2. Proposed Visual Navigation System Design

The proposed system is designed to estimate the drone's position and viewing direction using video input and telemetry data without relying on GNSS during real-time navigation. The system combines preprocessing techniques, visual feature extraction, and visual localization algorithms in order to estimate the drone trajectory and the ground point observed by the camera.
The proposed solution is divided into two main stages:

2.1. Preprocessing Stage

In the preprocessing stage, the system analyzes previously recorded drone flight data in order to build a visual reference database that will later be used during navigation. The preprocessing stage uses the drone videos together with the telemetry data extracted from the SRT files.
The first step is frame extraction from the drone videos. Frames are sampled periodically from the video stream and synchronized with the telemetry information such as GNSS coordinates, altitude, camera angle, and timestamp.
After synchronization, visual features are extracted from every frame using feature detection algorithms such as ORB (Oriented FAST and Rotated

BRIEF). These features allow the system to uniquely identify landmarks and visual patterns appearing in the environment.
For every processed frame, the extracted visual descriptors are stored together with their corresponding telemetry information inside a visual database. This database later serves as a reference map for real-time localization and navigation.

2.2. Real-Time Navigation Stage

During the navigation stage, the drone receives a real-time video stream without relying on GNSS positioning. The system continuously extracts visual features from the incoming frames and compares them with the previously created visual database.
Feature matching algorithms are used to identify similarities between the current frame and the stored reference frames from the preprocessing stage. Based on the matched visual features, the system estimates the camera pose and computes the approximate drone position and orientation.
Using the camera angle, altitude estimation, and the estimated camera pose, the system computes the approximate ground point located at the center of the camera view. Repeating this process continuously allows the system to estimate the drone trajectory in real time.

2.3. Selected Platform

For this project, ORB-SLAM3 was selected as the main visual navigation platform. ORB-SLAM3 is an open-source SLAM framework that supports monocular and visual-inertial navigation while providing accurate real-time localization and mapping capabilities.
The platform was selected because it provides:

- Real-time performance
- Robust feature-based tracking
- Support for drone navigation scenarios
- Loop closure detection
- Open-source implementation with extensive documentation

In this project, ORB-SLAM3 will be adapted to process the provided drone videos and estimate the drone trajectory using visual information extracted from the video stream.

3. Preliminary Experiment

In the preliminary experiment, the system is tested using one of the provided drone videos together with its corresponding telemetry (SRT) file. The purpose of the experiment is to estimate the drone trajectory and viewing direction using visual information and camera parameters extracted from the flight data.
The experiment focuses on validating the ability of the proposed system to estimate the drone path based on video analysis and compare it with the actual recorded trajectory extracted from the telemetry data.

3.1. Experiment Procedure

The experiment begins by extracting frames from the selected drone video at fixed time intervals. The telemetry information is extracted from the corresponding SRT file, including timestamp, altitude, GNSS coordinates, and camera angle.
Next, visual features are extracted from the video frames using ORB feature detection. Consecutive frames are matched in order to estimate the camera movement and approximate trajectory of the drone during flight.
The estimated trajectory is then compared with the actual trajectory recorded in the SRT telemetry file. The comparison allows evaluation of the system's ability to estimate the drone movement using visual information only.

3.2. Expected Results

The expected result of the experiment is a trajectory estimation that approximately follows the real flight path recorded by the drone telemetry system. Although small positioning errors are expected due to feature matching inaccuracies and environmental conditions, the estimated path should generally maintain the same flight direction and movement pattern. The experiment also demonstrates the feasibility of using visual navigation techniques for drone localization in GNSS-denied environments.

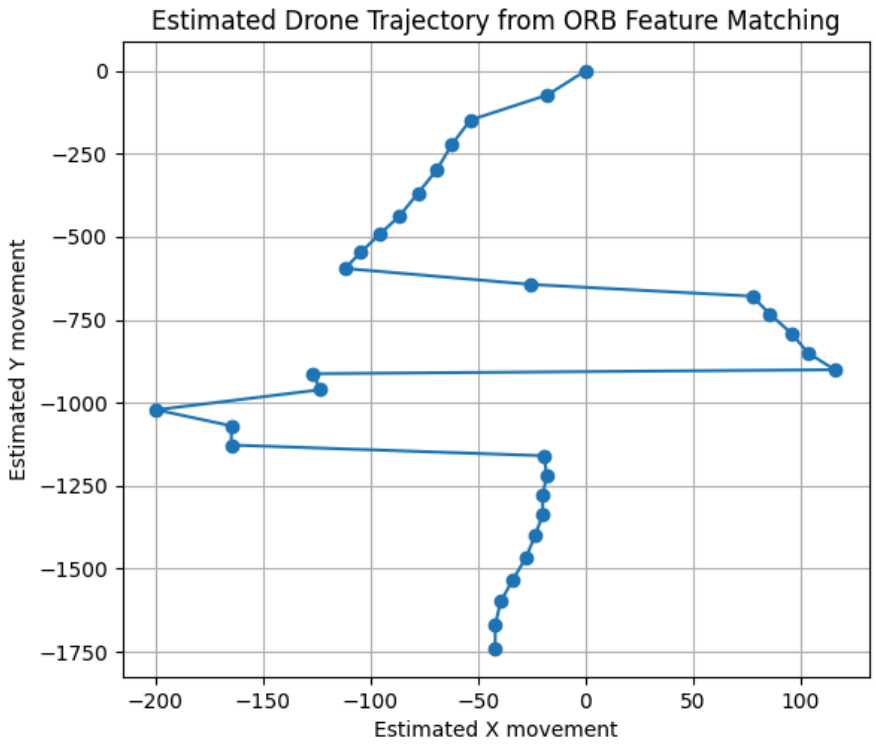

Figure 1: Estimated drone trajectory generated from ORB feature matching between consecutive video frames.
The preliminary implementation extracts frames from the selected drone video and applies ORB feature detection and matching between consecutive frames. The median displacement of matched features is accumulated over time to generate an approximate 2D trajectory. The result shows that visual features can be used to estimate the relative motion of the drone, although the trajectory is not yet geo-referenced and does not include telemetry correction.

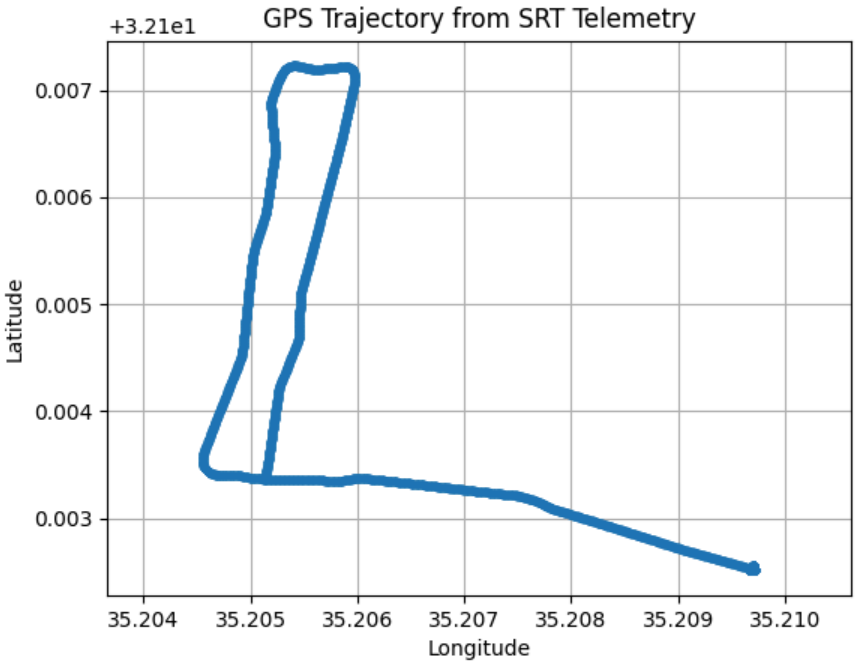

Figure 2: GPS trajectory extracted from the drone SRT telemetry file.
The GPS coordinates recorded by the drone telemetry system are visualized in order to represent the real flight path of the drone during the experiment.

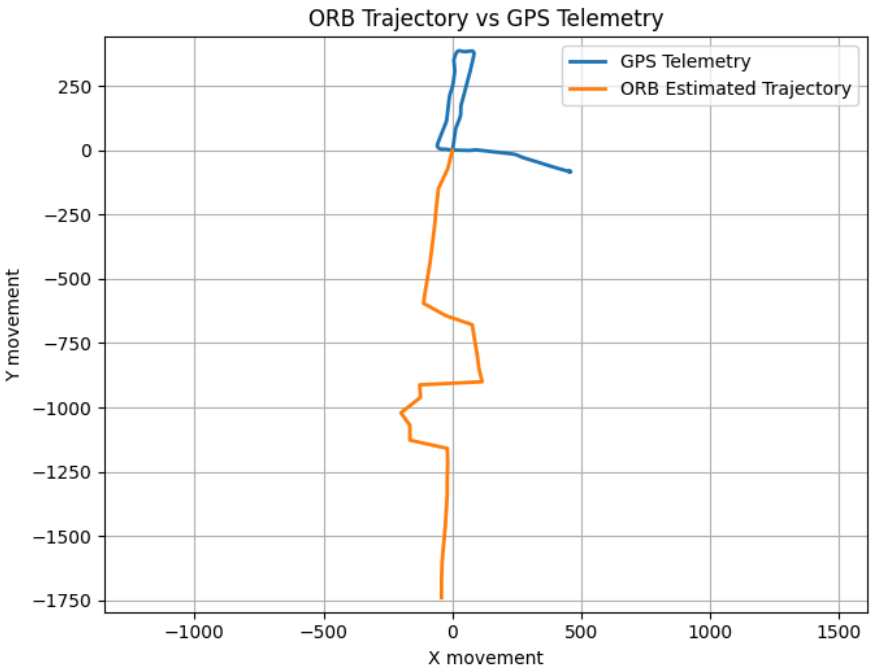

Figure 3: Comparison between ORB estimated trajectory and GPS telemetry trajectory.
The comparison demonstrates that the visual trajectory estimation approximately follows the overall movement trend of the real drone flight path, although scale and orientation differences still exist.

3.3. Simulated Real-Time Localization

A simulated real-time localization module was implemented in order to estimate the drone position from an incoming video frame. A visual reference database was built from preprocessed frames extracted from the drone video together with their associated telemetry data.
During localization, ORB features are extracted from the query frame and matched against the reference database. The GPS coordinates associated with the best matching frame are returned as the estimated drone position. This experiment demonstrates the feasibility of performing visual localization without direct GNSS measurements during real-time navigation.

4. Future Work

Future improvements may include integrating inertial sensor data (IMU), applying deep-learning-based feature extraction methods, improving trajectory optimization, and testing the system under more challenging conditions such as nighttime environments, rapid motion, and low-texture scenes.
Future improvements may also include full scale and rotation alignment between the visual trajectory and the GPS telemetry in order to achieve more accurate geo-referenced localization.

5. Conclusion

This work presented a preliminary design for a drone visual navigation system based on visual SLAM and feature-based localization techniques. Several modern visual navigation frameworks and algorithms were reviewed and compared, including visual-inertial odometry, SLAM, and deep-learning-based approaches.
The proposed system combines preprocessing, feature extraction, and real-time localization methods in order to estimate drone trajectory without relying on GNSS during navigation. The preliminary experiment demonstrates the feasibility of using visual navigation techniques for autonomous drone localization and trajectory estimation.